Rotary Wing Aerodynamics (AE 667)

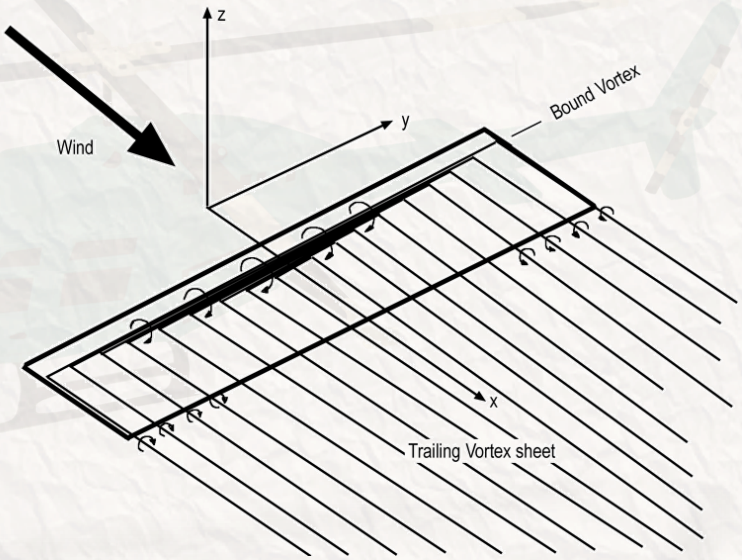


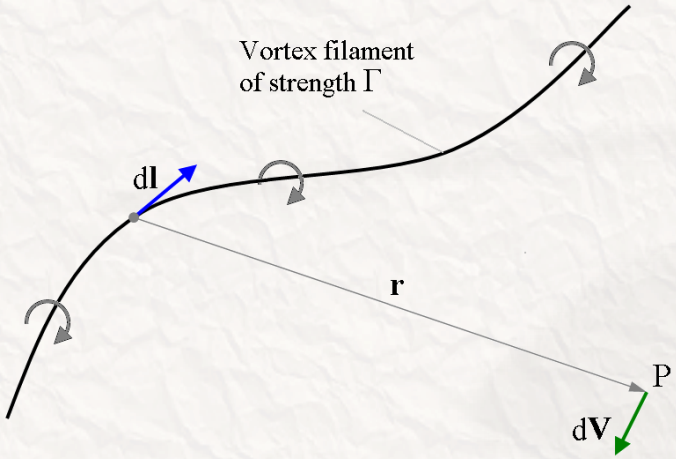
Vortex Theory

Background

- Based on Prandtl's lifting line theory

- Uses:
  - Rotor wake models
  - Kutta–Joukowski theorem
  - Biot-Savart Law





$$L' = -\rho_\infty V_\infty \Gamma$$

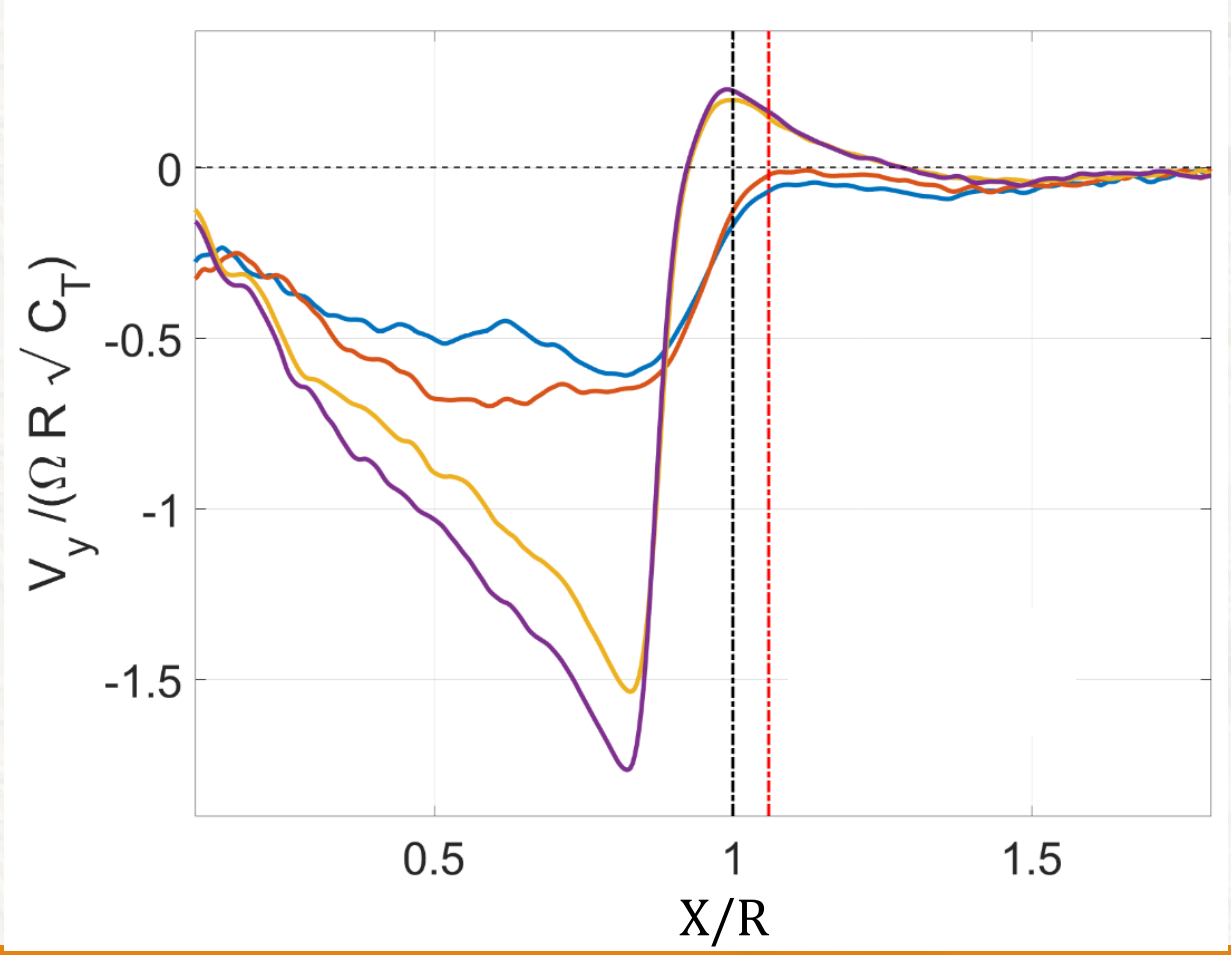
Background

Addresses predictions problems at high thrust conditions (high C_T/σ)

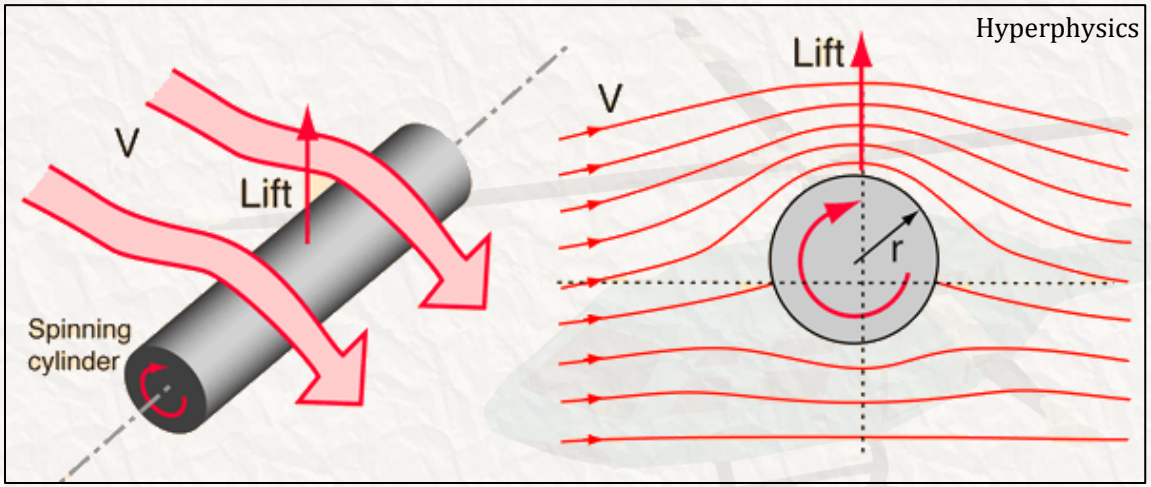
- High wake contraction at high thrust condition
- Upward directed inflow!
- Affecting tip loading and power consumption



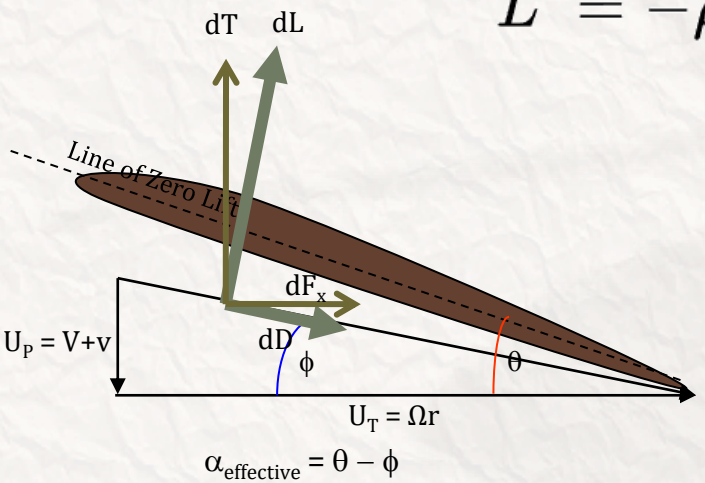
Some experimental results



Kutta-Joukowski Theorem



$$L' = -\rho_\infty V_\infty \Gamma$$

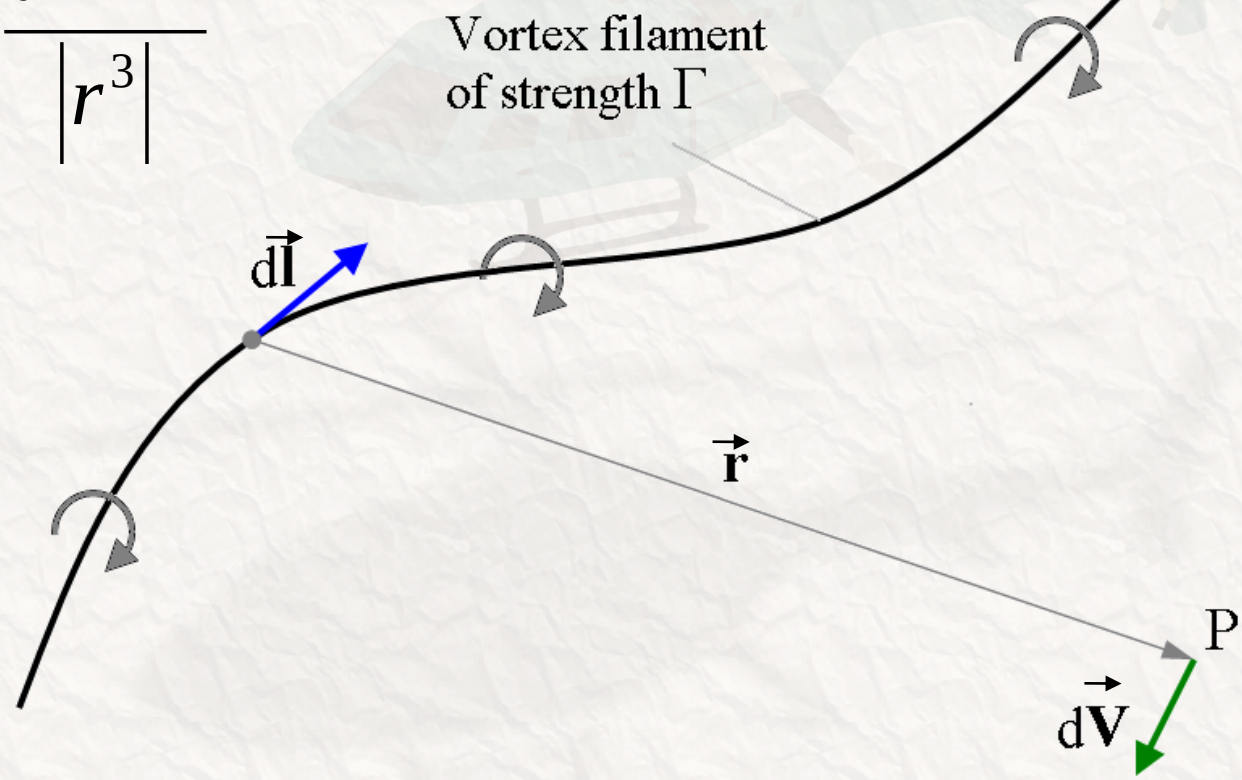


$$dT = \rho(\Omega r)\Gamma$$
$$dF_x = \rho(V+v)\Gamma$$

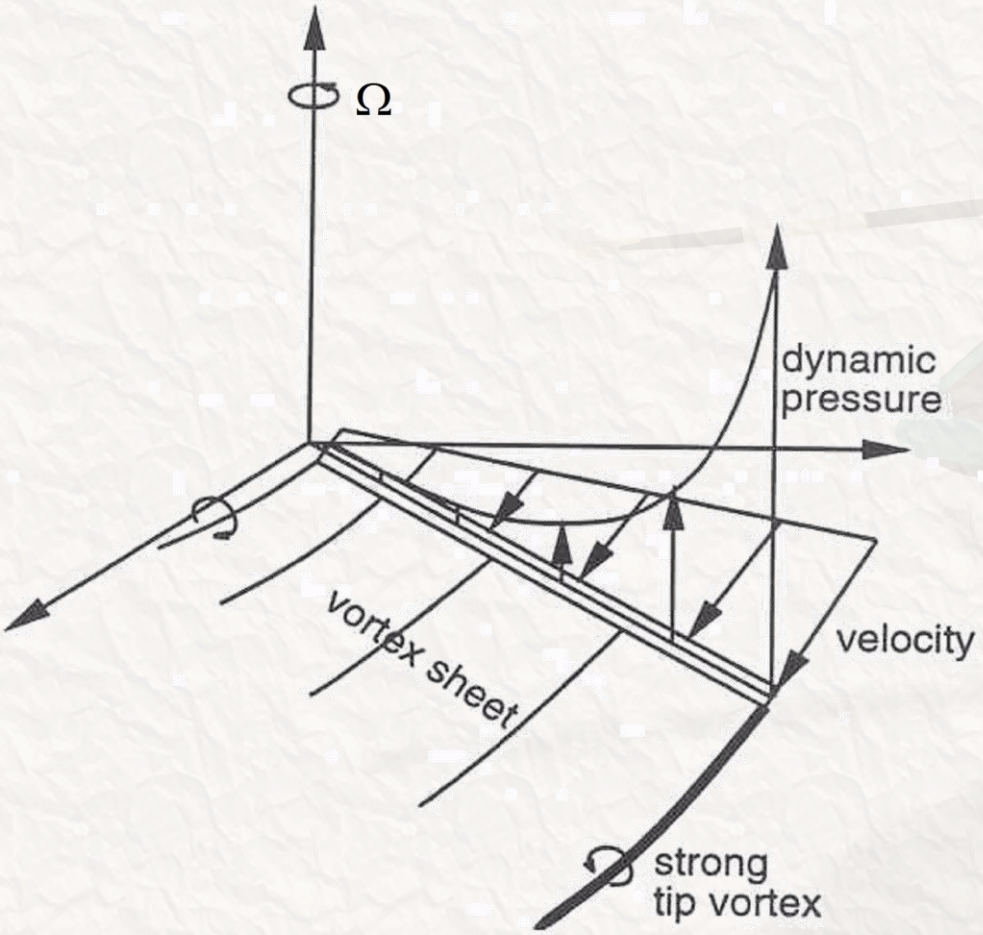
Where **Γ** is the bound circulation at the airfoil section

Biot-Savart Law

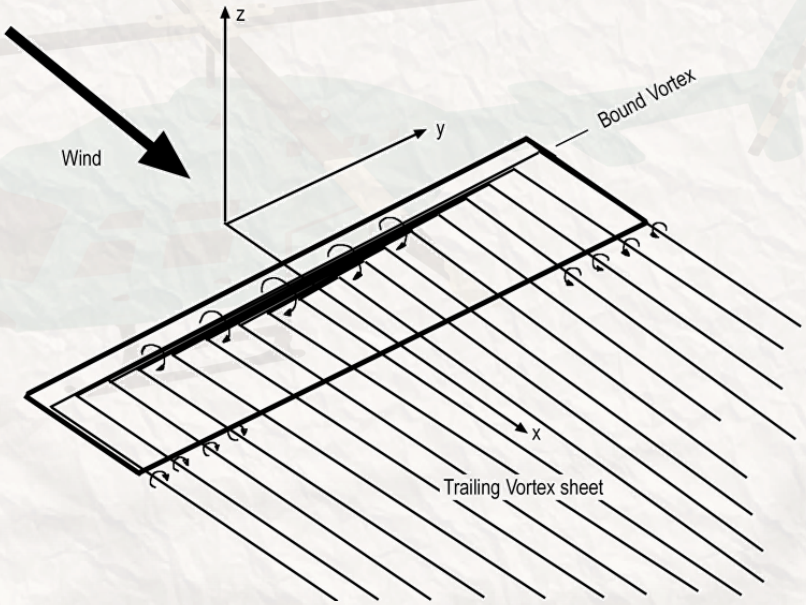
$$d\vec{V}_{induced} = \frac{\Gamma}{4\pi} \frac{d\vec{l} \times \vec{r}}{\left| r^3 \right|}$$



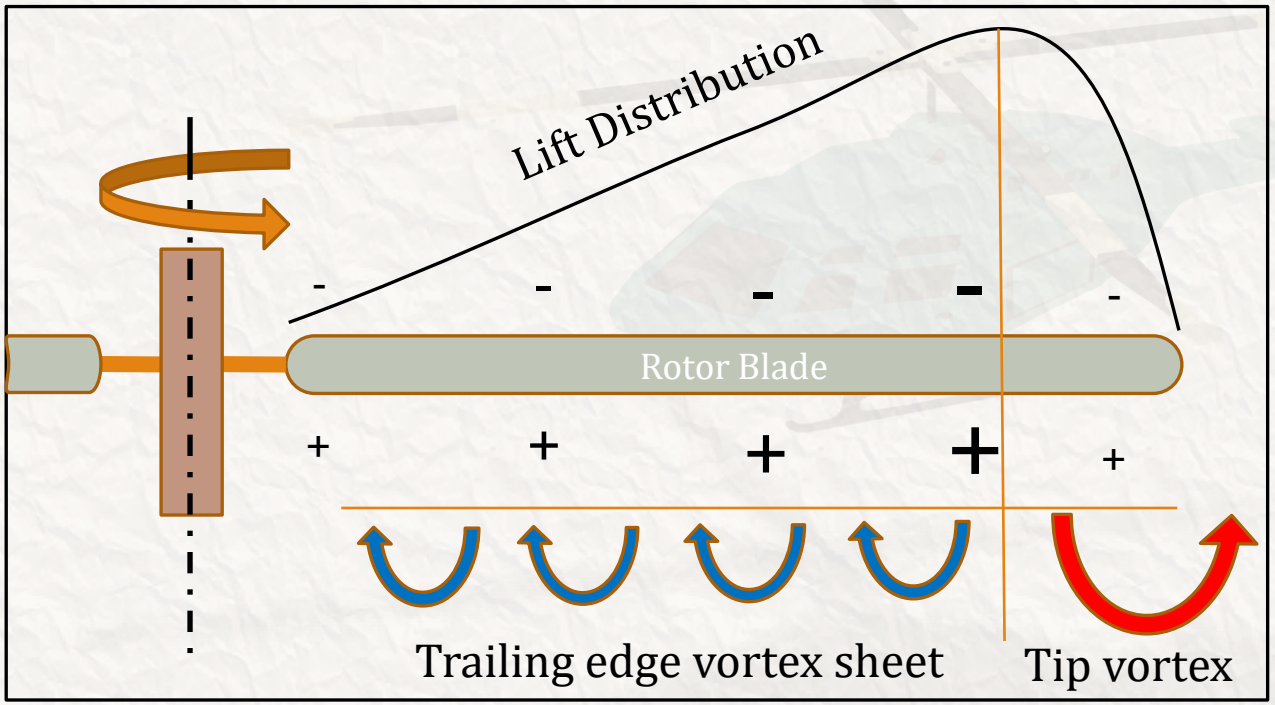
Vortices in Rotor Wake



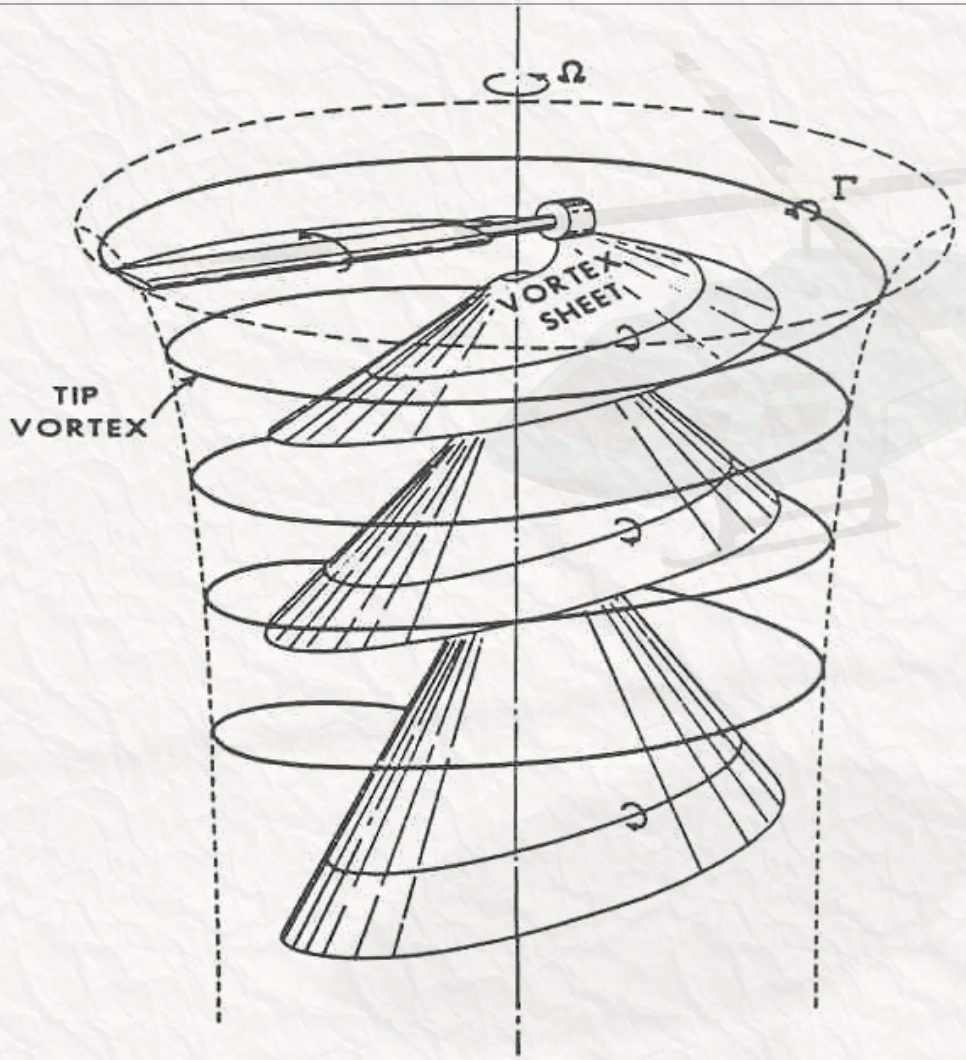
Rotor



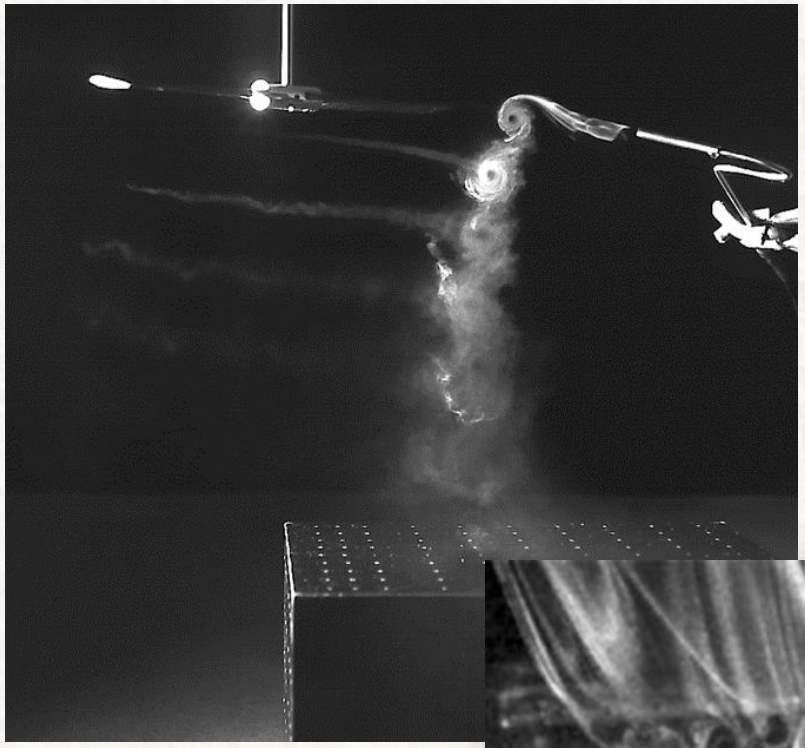
Fixed Wing

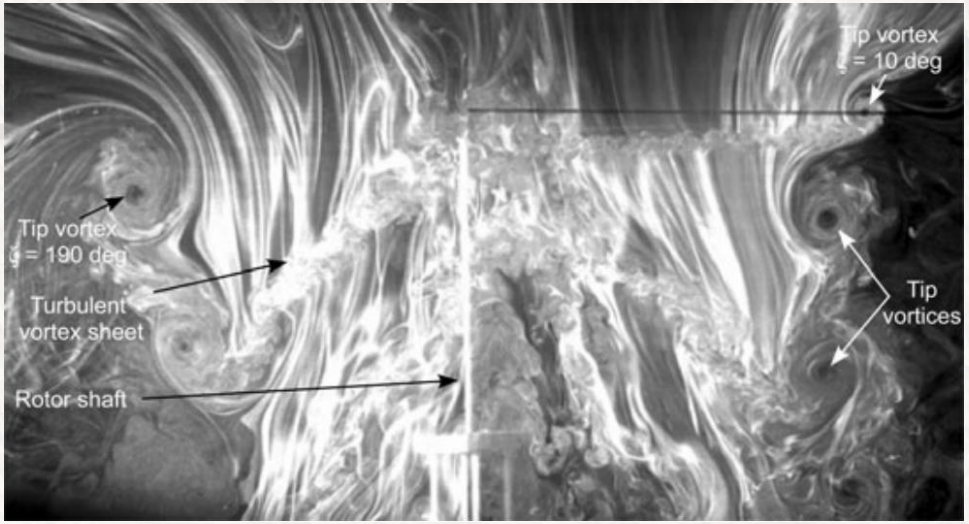

Lift Distribution
Rotor Blade
Trailing edge vortex sheet
Tip vortex

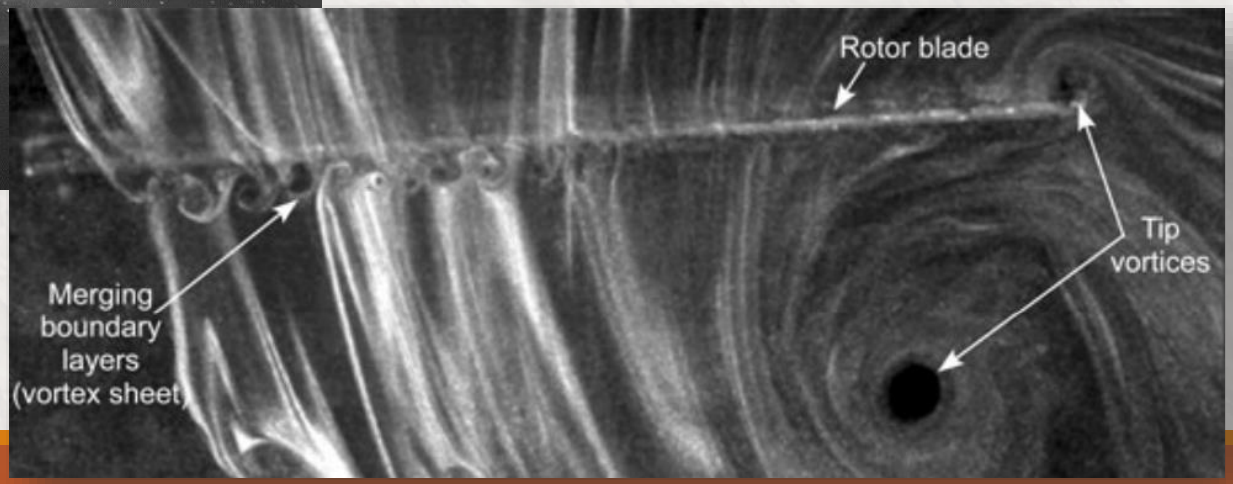
Prescribed Wake Model (Robin Gray)



Some Experimental Results

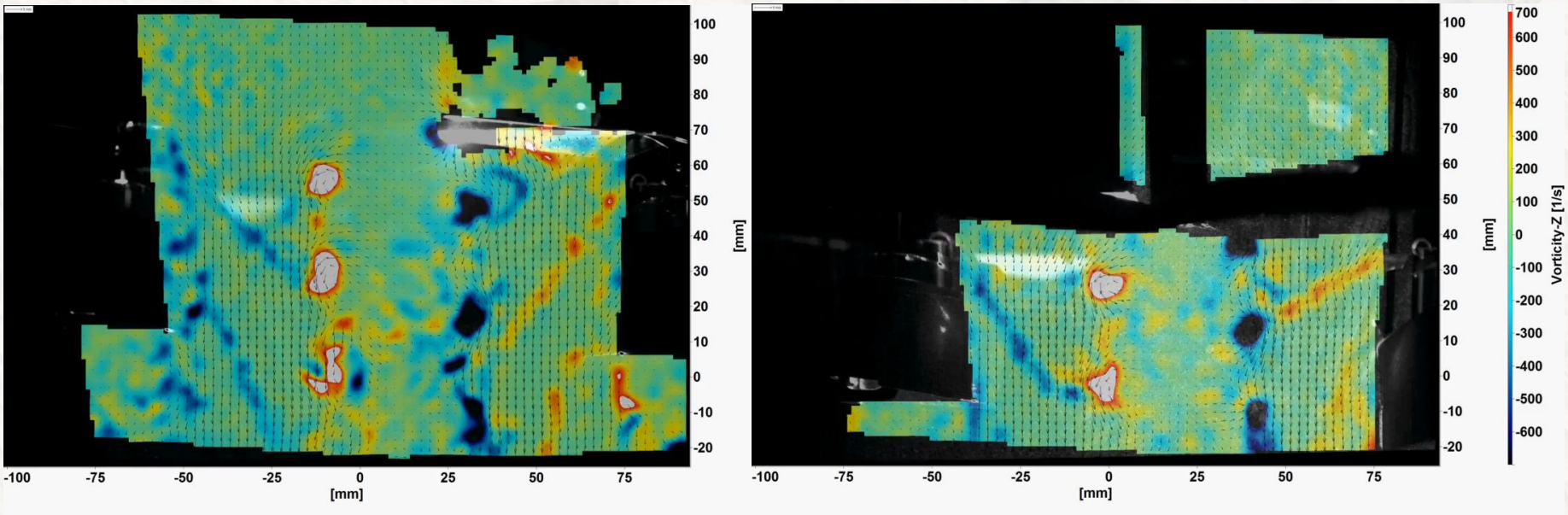






Ramasamy, M., Johnson, B., & Leishman, J. G. (2008). Understanding the Aerodynamic Efficiency of a Hovering Micro-Rotor. *Journal of the American Helicopter Society*, *53*(4), 412-428.

Some Experimental Results



Prescribed Wake Model (Landgrebe's Curve Fit)

Radial position of the tip vortex :

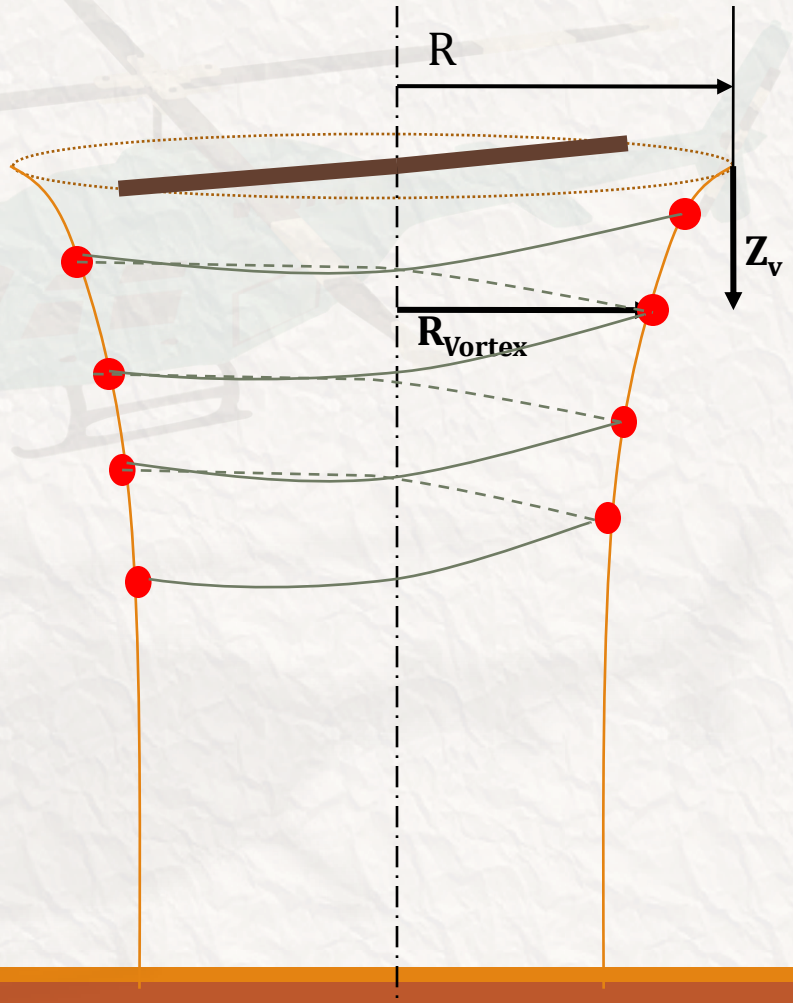
$$\frac{R_{vortex}}{R} = A + (1 - A)e^{-\lambda \psi_v}$$

$$A = 0.78$$

$$\lambda = 0.145 + 27C_T$$

ψ_v = Vortex Age

= Azimuthal Position of the vortex Filament measured from the blade

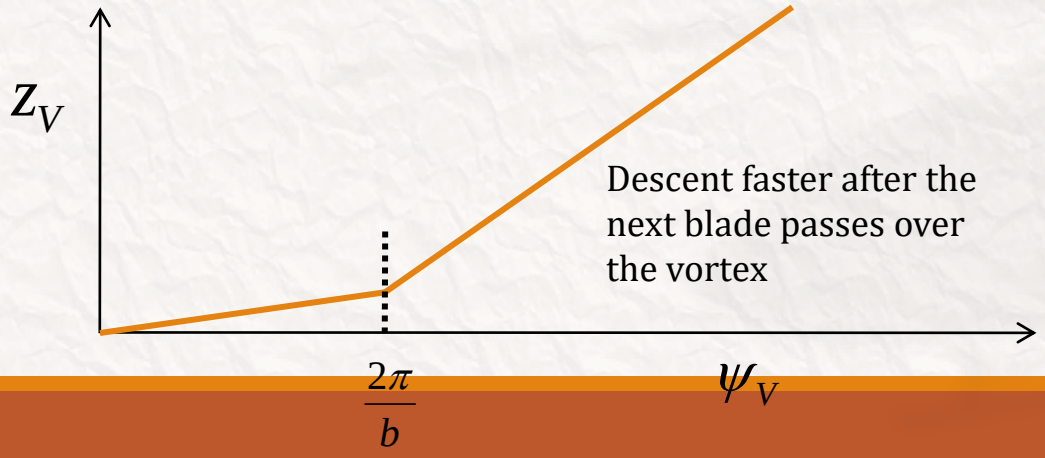


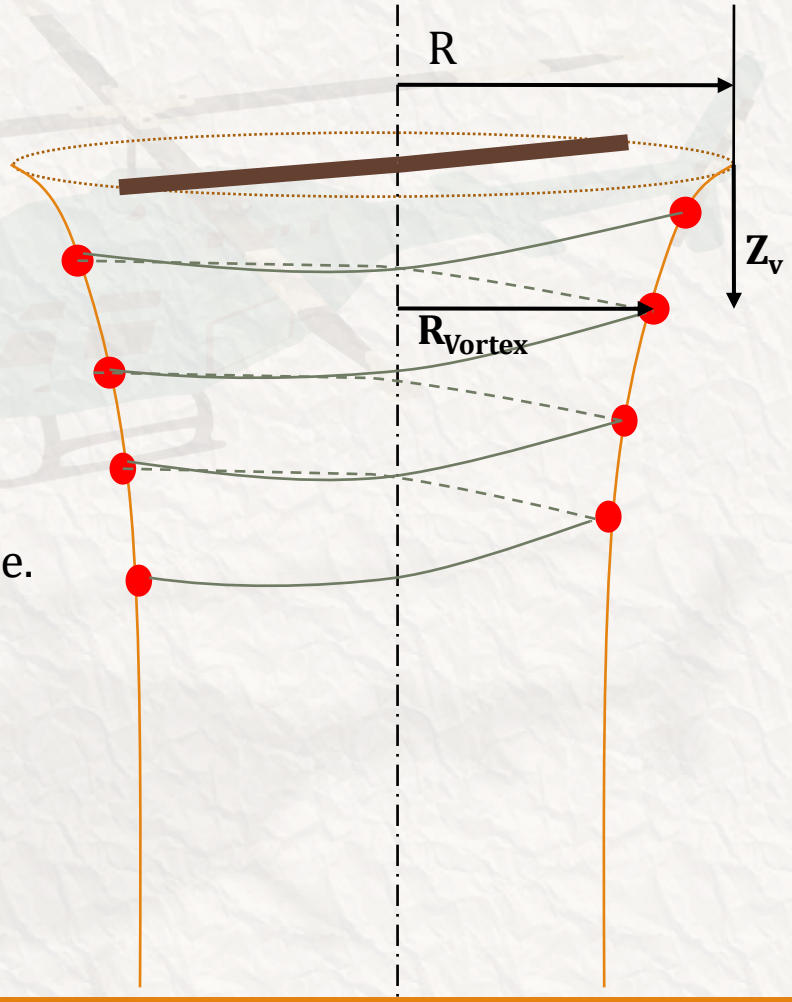
Prescribed Wake Model (Landgrebe's Curve Fit)

- $\frac{z_V}{R} = k_1 \psi_V \qquad 0 \le \psi_V \le \frac{2\pi}{b}$
- $\frac{z_V}{R} = k_1 \frac{2\pi}{b} + \mathrm{k}_2 \left(\psi_V - \frac{2\pi}{b}\right) \qquad \psi_V \ge \frac{2\pi}{b}$
- $k_1 = -0.25 \left[\frac{C_T}{\sigma} + 0.001\theta_{\text{twist, degrees}}\right]$
- $k_2 = -\sqrt{C_T} - 0.01\sqrt{C_T}\theta_{\text{twist, degrees}}$

Here, $\theta_{\text{twist,degrees}}$:

Blade twist = Tip Pitch angle – Root Pitch Angle.

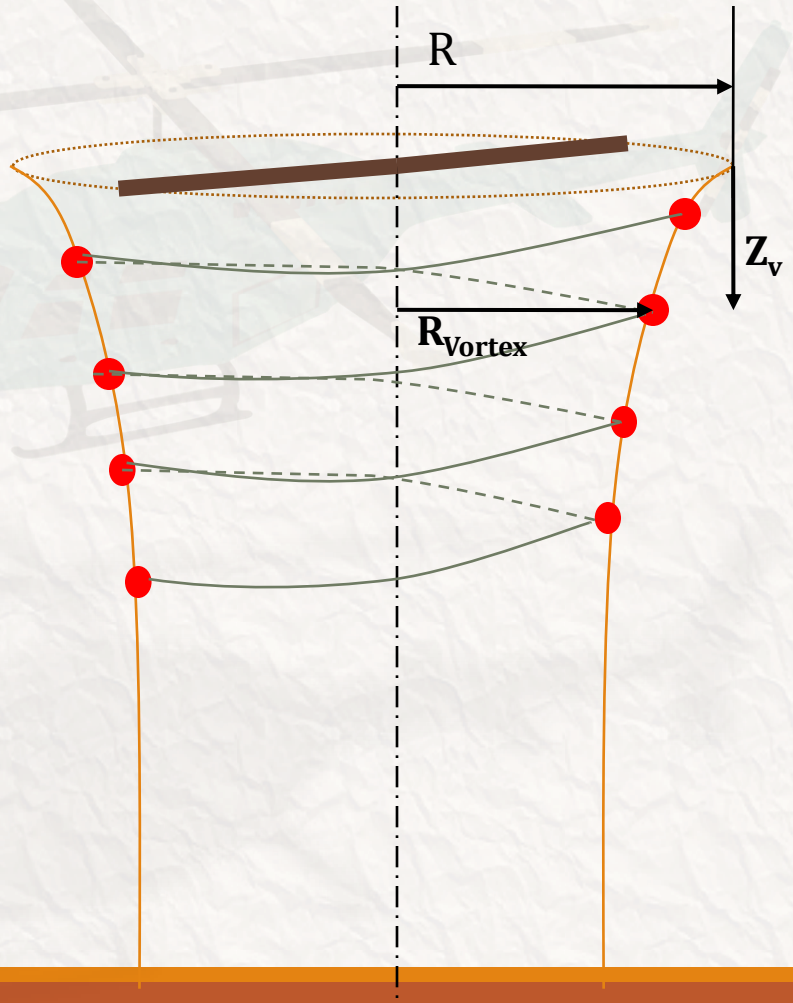




Prescribed Wake Model (Landgrebe's Curve Fit)

Landgrebe improvised the model later (1977).

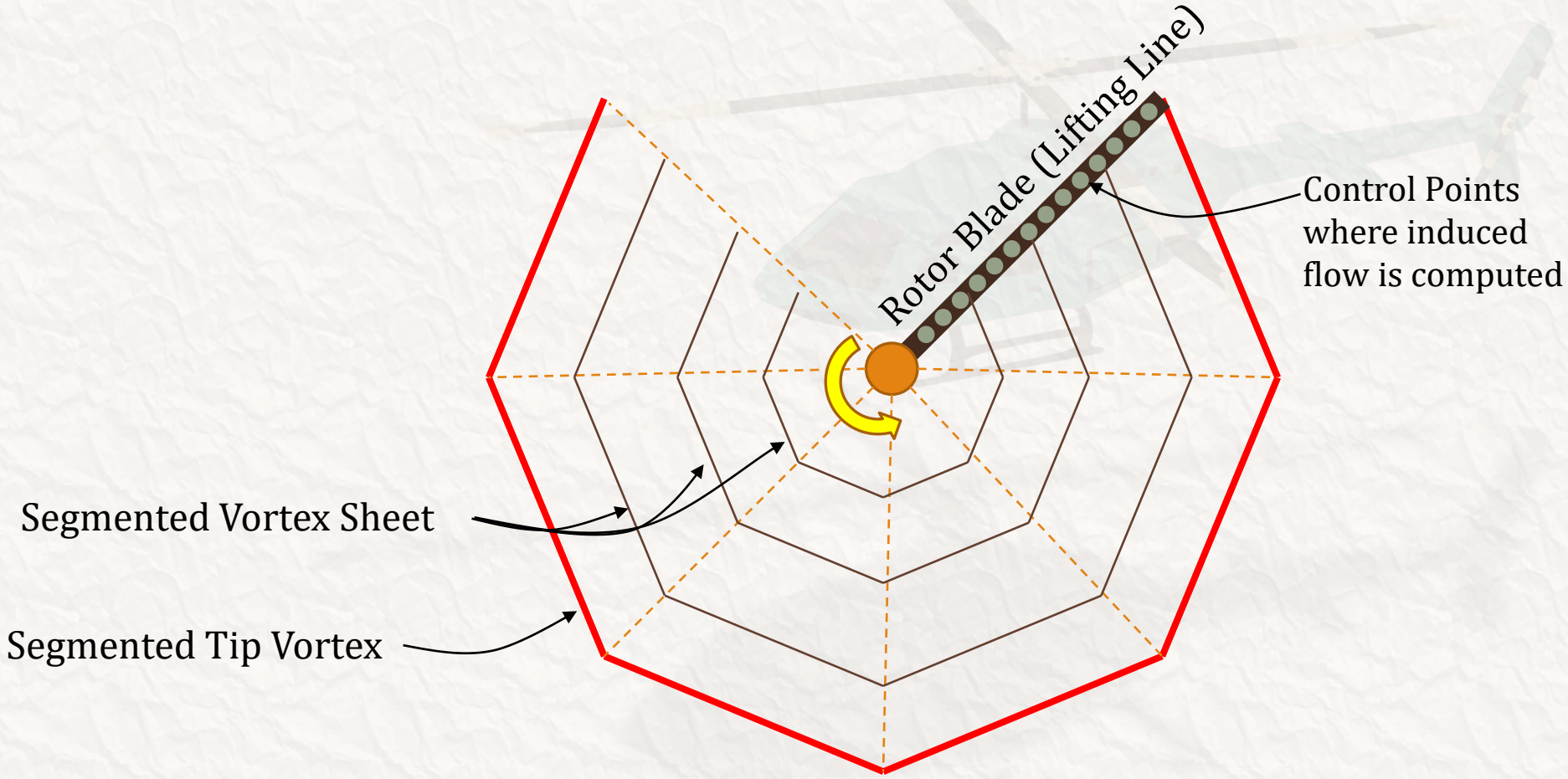
Based the trajectory on **peak bound circulation** instead of C_T/σ.



Using vortex models for computing

- Continuous helical tip vortex structure broken into piecewise straight line segments.
- Tip vortex strength assumed to be about 80% of max bound circulation
- Vortex core of empirically prescribed radius defined for keeping induced velocities finite.

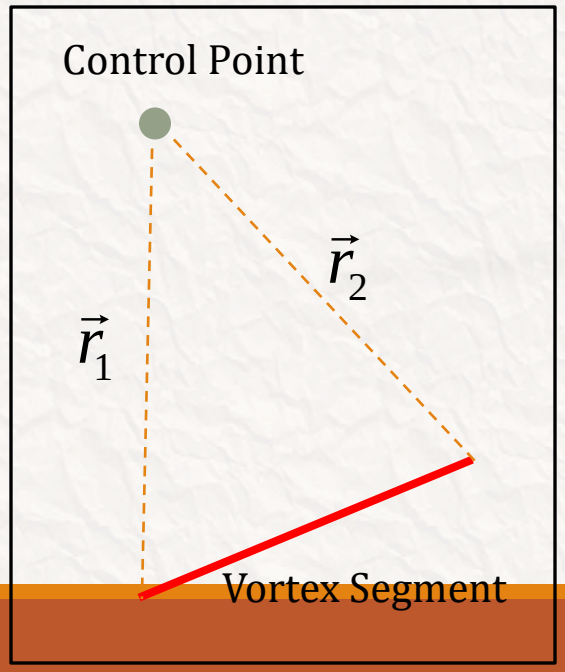
Using vortex models for computing



Using vortex models for computing

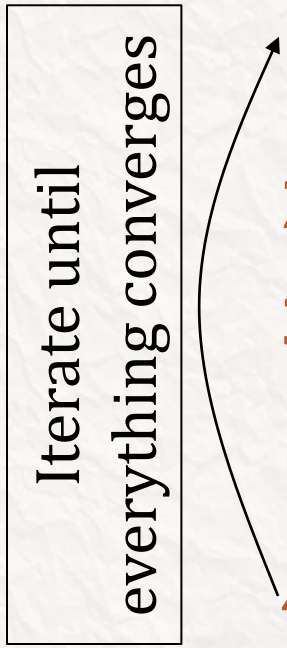
Using Biot - Savart Law

$$\vec{V}_{induced,control} = \frac{\Gamma}{4\pi} \frac{(\vec{r}_1 \times \vec{r}_2)(r_1 + r_2)\left(1 - {\vec{r}_1 \bullet \vec{r}_2}/{r_1 r_2}\right)}{(r_1 r_2)^2 - (\vec{r}_1 \bullet \vec{r}_2)^2 + r_c^2 (r_1^2 + r_2^2 - 2\vec{r}_1 \bullet \vec{r}_2)}$$



Here, r_c is the vortex core radius

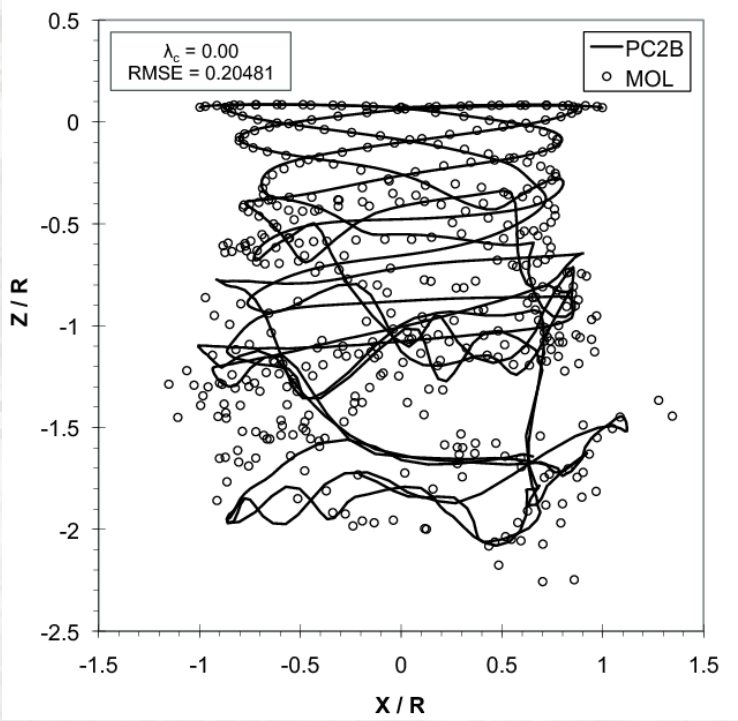
Pseudo code



1. Compute inflow using Biot-Savart law (BEM for the first iteration)
2. Compute radial lift distribution
3. Find circulation strengths from lift distribution. 80% of the peak circulation is tip vortex strength
4. Assume prescribed wake trajectory (Landgrebe's, or any other)

Free Wake Model

- No empirical wake model.
- Vortex segments allowed to get influenced by velocities induced by other segments and move freely in space
- Wake shape (position of vortex segments) updated at each time segment



Betoney, Patrick et al. "Development of a free-vortex wake model in state-space form." (2009).